

TRIASHA SARKAR

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Machine Learning Engineer and Georgia Tech aerospace MS graduate who builds dependable systems from data and experiments through deployment. Experience spans retrieval and ranking, scientific ML, evaluation, distributed training, and high-throughput serving.

EDUCATION

MS, Aerospace Engineering Georgia Institute of Technology, Atlanta, GA	Aug 2026
Entered PhD program; transitioned to MS, completed Aug 2026	
MSc, Aerospace Vehicle Design Cranfield University, United Kingdom	Feb 2023
First Class Honours · Thesis on geometry optimization under FAA constraints	
B.Tech, Aerospace Engineering SRM Institute of Science and Technology, India	Jun 2021
First Class Honours	

TECHNICAL SKILLS

ML & GenAI: PyTorch, FSDP, DeepSpeed, Megatron-LM, Transformers, vLLM, SGLang, TensorRT-LLM, RAG, hybrid retrieval, reranking, PEFT/LoRA, GRPO evaluation

Systems: Python, Go, SQL, Kafka, FastAPI, Docker, Kubernetes, GitHub Actions, PostgreSQL/pgvector, Prometheus, OpenTelemetry, ONNX, Core ML, Qualcomm QNN

Scientific computing: NumPy, SciPy, pandas, NetworkX, OSMnx, statistical inference, design of experiments, surrogate modeling

PROFESSIONAL EXPERIENCE

Graduate Research Assistant under Prof. Dimitri Mavris Georgia Tech ASDL	May 2025 – Aug 2026
• Took simulation, safety, and sustainability questions from data review and model design through tested analysis, with reproducible evidence for faculty and sponsors. • Checked assumptions with domain experts, documented limitations, and delivered models and workflows that others could rerun and use.	
Machine Learning Engineer Rolls-Royce	Jul 2023 – Apr 2025
• Built Python workflows for diagnostics, anomaly detection, predictive maintenance, and mixed-frequency time series from multivariate engine data. • Turned certification requirements into traceable analyses and model checks, reviewing each result with lifecycle engineers before use.	
Data Science Intern Rolls-Royce DataLabs	May – Jul 2021
• Cleaned aircraft-engine sensor data, designed features, and compared predictive and anomaly-detection models to identify recurring failure patterns.	

SELECTED TECHNICAL PROJECTS

AeroRAG-X Retrieval, Post-Training, and ML Systems	2025 – Present
github.com/triasha72/AeroRAG-X • Built hybrid BM25/dense retrieval, RRF, cross-encoder reranking, pgvector search, evidence gating, and citation controls over 3,233 source-preserving NASA NTRS chunks. • Added protected Base/LoRA evaluation, bounded agents, FastAPI services, observability, distributed-training treatments, and high-throughput serving paths. GPU result tables remain pending.	
IntegrityBench Policy-Conditioned Moderation Evaluation	Aug 2026 – Present
github.com/triasha72/IntegrityBench • Built a 360-case benchmark across policy shifts, long context, multilingual, adversarial, standard, and low-resource slices with structured allow/reject/escalate decisions. • A static-policy control reached 0.620 macro F1 and 0.286 false acceptance; current-policy retrieval reached 1.000 on the controlled generator. Reported as a construction check, not production evidence.	
NewsLens Recommendation, Real-Time Search, and Production ML	Jul – Aug 2026
github.com/triasha72/NewsLens • Used chronological splits, TF-IDF ranking, and a training-only fallback for leakage-safe MIND recommendation, reaching NDCG@10 of 0.366; the one-time holdout interval included zero. • Built a Go, Kafka, PostgreSQL, and FastAPI article path with idempotency and dead letters. A 500-event local run measured 44 ms publish p99, 79 ms freshness p95, and 5.7 s partition recovery.	
EdgeGenBench Scientific ML and On-Device Inference	Aug 2026 – Present
github.com/triasha72/EdgeGenBench • Built an aircraft-design surrogate benchmark with uncertainty estimates, constrained optimization, physics checks, ONNX export, and deployment tests. • The Qualcomm QNN model retained 0.997 mean held-out R²; a drifting INT8 candidate was rejected by policy. Shipped an installable Safari app without claiming native-device performance.	

AeroSynth-Eval | Multimodal AI Evaluation

Aug 2026 – Present

github.com/triasha72/AeroSynth-Eval

- Built a fixed 48-scenario VLM suite with deterministic assets, provenance checks, annotation workflows, and typed evaluator contracts.
- A Kaggle P100 batch attempted all 12 development cases: eight passed the response schema and four JSON failures were retained for diagnosis.

Silent Failure Detection in Rocket-Motor Simulations

May – Jul 2026

Georgia Tech ASDL

- Audited a design sweep that failed without errors, built a leakage-safe classifier with 96.8% accuracy across 15,120 simulations, and traced the failures to an uncapped convergence loop.
- Tested unseen geometries separately and wrote a patch that converts silent hangs into logged convergence errors.

HERO | Source-Aware Retrieval for Airline Safety

2025 – 2026

Georgia Tech ASDL

- Worked on retrieval and evaluation methods that keep generated safety answers tied to the technical sources analysts need to review.

GREEN TEA and Project EAGLE | Sustainable Aviation Modeling

Sep 2025 – Apr
2026

Georgia Tech ASDL

- Built surrogate, uncertainty, demand, and life-cycle models for alternative-fuel and transport studies; the GREEN TEA model remains in sponsor use.

Atlanta Mobility Resilience Digital Twin

2026 – Present

github.com/triasha72/atlanta-mobility-resilience-digital-twin

- Built an OSMnx and NetworkX simulator for road-closure effects on Atlanta travel time and access, with deterministic core tests in CI and live OSM download kept as an explicit integration step.

Equity Backtest | Walk-Forward Signal Evaluation

Jul 2026

github.com/triasha72/Equity-Backtest

- Built an expanding-window backtest with transaction costs and retained every specification tested; the apparent momentum edge disappeared after costs and missed the preregistered significance threshold.

LEADERSHIP

Graduate Chair | Women of Aeronautics and Astronautics

2025 – 2026

Georgia Tech